Write

$$P(x) = x^7 + a_6x^6 + \cdots + a_1x + a_0$$

and set

$$A = |a_0|.$$

By hypothesis, A is a positive odd integer less than 2025^2, and $P(A) = 0$.

Since A is an integer root and P is monic,

$$P(x) = (x - A)Q(x)$$

for some monic $Q \in \mathbb{Z}[x]$ of degree 6. Moreover,

$$Q(0) = -\frac{a_0}{A} \in -1, 1.$$

Now let b be the absolute value of any coefficient of P. Then $P(b) = 0$. If $b \neq A$, then

$$0 = P(b) = (b - A)Q(b),$$

so $Q(b) = 0$. Since b is a positive integer root of the monic polynomial Q, the rational-root theorem gives

$$b \mid Q(0).$$

But $|Q(0)| = 1$, hence $b = 1$.

Therefore every coefficient of P has absolute value either 1 or A.

0.0.1 Case 1: $A = 1$

Then every coefficient has absolute value 1. Thus

$$a_0, a_1, \ldots, a_6 \in -1, 1.$$

The only coefficient magnitude is 1, so the root condition is exactly $P(1) = 0$.

Suppose exactly r of the seven coefficients $a_0, \ldots, a_6$ equal 1. Then

$$P(1) = 1 + r - (7 - r) = 2r - 6.$$

Thus $P(1) = 0$ precisely when $r = 3$. Hence this case contributes

$$\binom{7}{3} = 35$$

polynomials.

0.0.2 Case 2: $A > 1$

Since A is odd, $A \geq 3$. Write

$$a_i = \varepsilon_i A^{\delta_i}, \qquad \varepsilon_i \in -1, 1, \quad \delta_i \in 0, 1.$$

Because $|a_0| = A$, we have $\delta_0 = 1$.

We use the following elementary observation: if $A \geq 3$, $|c_j| \leq 2$, and

$$\sum_{j=0}^{n} c_j A^j = 0,$$

then every $c_j = 0$. Indeed, if n is the largest index for which $c_n \neq 0$, then

$$|c_n| A^n \geq A^n,$$

whereas

$$\left| \sum_{j=0}^{n-1} c_j A^j \right| \leq 2 \sum_{j=0}^{n-1} A^j = \frac{2(A^n - 1)}{A - 1} \leq A^n - 1,$$

a contradiction.

Now

$$0 = P(A) = A^7 + \sum_{i=0}^{6} \varepsilon_i A^{i+\delta_i}.$$

After collecting equal powers of A, every resulting coefficient has absolute value at most 2. Hence every such coefficient must vanish.

Starting with the highest powers gives successively:

$$\delta_6 = 1, \qquad \varepsilon_6 = -1,$$

$$\delta_5 = 0,$$

$$\delta_4 = 1, \qquad \varepsilon_4 = -\varepsilon_5,$$

$$\delta_3 = 0,$$

$$\delta_2 = 1, \qquad \varepsilon_2 = -\varepsilon_3,$$

$$\delta_1 = 0,$$

$$\delta_0 = 1, \qquad \varepsilon_0 = -\varepsilon_1.$$

Consequently, for some $s, t, u \in -1, 1$,

$$P(x) = x^7 - Ax^6 + sx^5 - sAx^4 + tx^3 - tAx^2 + ux - uA \quad = (x - A)\big(x^6 + sx^4 + tx^2 + u\big).$$

The leading coefficient has absolute value 1, so 1 must also be a root. As $A \neq 1$,

$$0 = P(1) = (1 - A)(1 + s + t + u),$$

and therefore

$$1 + s + t + u = 0.$$

Thus

$$s + t + u = -1,$$

which means exactly one of s, t, u equals 1 and the other two equal -1. There are therefore exactly

$$3$$

polynomials for each admissible $A > 1$.

Conversely, every polynomial of this form with exactly one of s, t, u positive has coefficient magnitudes 1 and A, and both 1 and A are roots, so all such polynomials are indeed good.

Finally,

$$2025^2 = 4,100,625.$$

The number of odd integers A satisfying

$$1 < A < 4,100,625$$

is

$$\frac{4,100,625 - 3}{2} = 2,050,311.$$

Hence the total number of good polynomials is

$$35 + 3(2,050,311) = 35 + 6,150,933 = \boxed{6,150,968}.$$